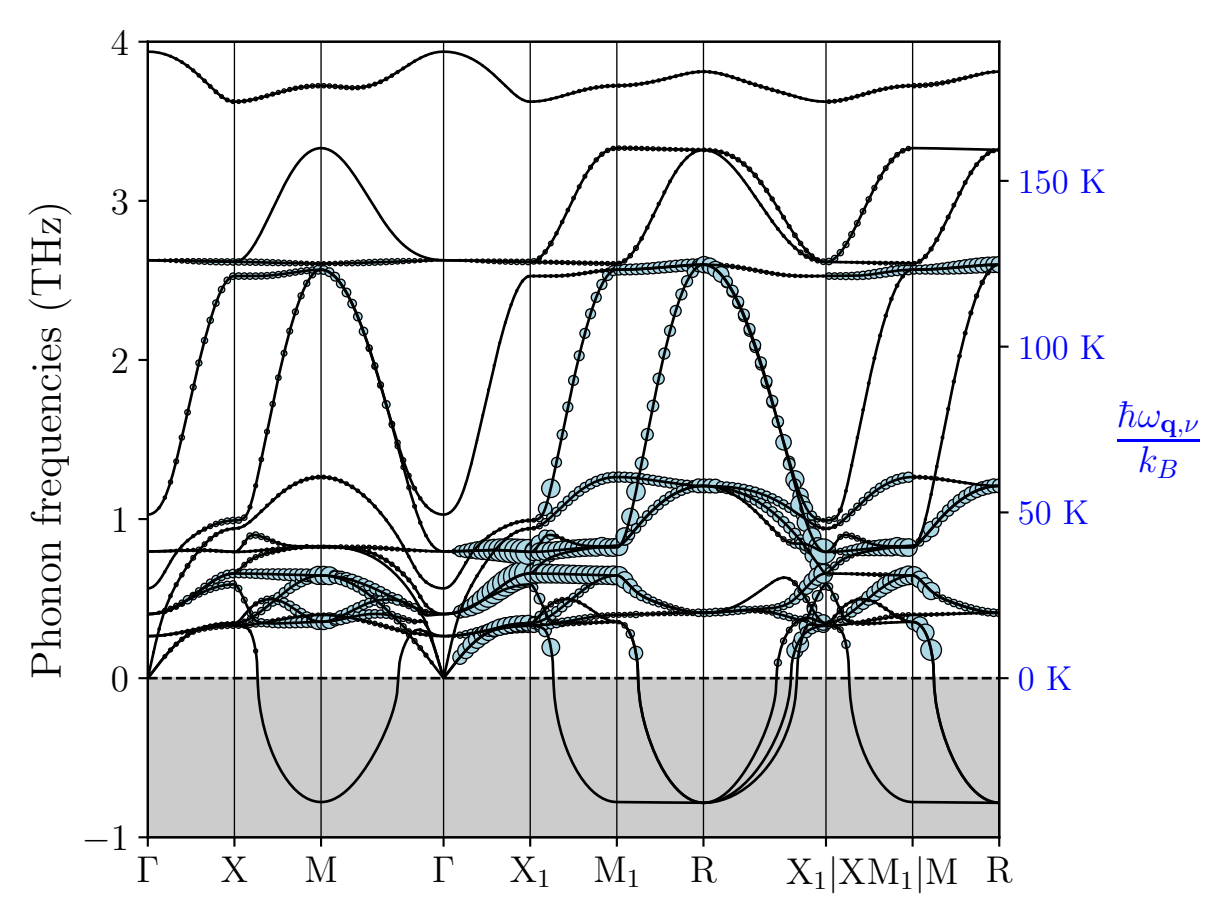

Phonon frequencies (THz)
4
3
2
1
0
−1
Γ
X
M
Γ
X1
M1
R
X1|X
M1|M
R
150 K
100 K
50 K
0 K
ħω_{q,ν} / k_B